Treetable

Receipted ecological sensing.

Every field reading carries a replayable evidence chain.

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Problem — nature monitoring can't be audited

- Biodiversity credits, MRV, and conservation claims are under integrity scrutiny — buyers increasingly ask "show me the evidence," and vendors hand back PDFs.
- Sensing hardware is expensive ($600+) or proprietary; the data pipeline is a black box.
- Meanwhile, "biological compute" vendors (wetware labs) sell results that literally cannot be replayed or verified.

Monitoring is the bottleneck for nature action — you can't credit what you can't verifiably measure.

Solution — an evidence layer for sensing

- **Receipts, not reports.** Every reading is content-addressed (SAIDed), witnessed by a quorum of independent nodes, and replayable by any auditor.
- **Consent-gated by design.** Emit lanes require joint guardian+operator consent; refused lanes are first-class values — biological subjects are participants, not inputs.
- **Replay-verified compute.** Deterministic inference over 15 real EM-derived organism connectomes — every task ships a trace digest any caller can reproduce bit-for-bit.

Product — three surfaces, one stack

- **EcoCell One** — universal $35–45 field node (ESP32-S3, multimodal sensor heads, LoRa/Reticulum mesh relay). Manufacturable KiCad, DRC-clean, assembly-ready.
- **Evidence layer** — AGPL open source: sense/fleet lanes, quorum → receipt → mutual-credit settlement. 90/90 e2e checks.
- **Compute API** — live today: connectome-compute-api.terexmaps.workers.dev. Deterministic LIF over real wiring, receipts on every call.

BYOD path: the evidence layer runs on phones/Pis — the network doesn't wait on hardware.

Why the evidence model wins

- Every claim is tiered: **measured** / estimated / cited
- Controls built in — a degree-matched shuffled-topology control refuted our own flattering interpretation, and we published it
- Real RAPL energy measurements, replay-verified bundles

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medulla (40k neurons):
connectome   1.17e-1 J
dense fp32   8.30   J   (~70×)
sparse fp32  2.11e-1 J

→ sparse wiring beats dense O(n²)
  at scale — measured, not claimed
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Buyers of verification buy the methodology, not the marketing.

Traction (Oct 2026)

- Open-source stack live: sensing→quorum→receipt→settlement, 90/90 e2e checks, AGPL
- **Compute API deployed** — replay-verified tasks over 15 connectomes, cross-implementation digests identical (930/930)
- EcoCell One carrier PCB: fab-ready Gerbers, BOM, pick-and-place files
- NSF SBIR/STTR Project Pitch submitted (#00131003, Environmental Technologies)
- Crowd Supply hardware campaign application submitted
- Benchmark: first measured efficiency table over real connectomes with honesty tiers

Business model — zero-capital path to revenue

- **Now:** verification audits for wetware/sensing vendors ($7.5–25k engagements — Cortical Labs, FinalSpark contacted), benchmark certification reports ($2.5–10k), replay-verified compute API (free → $99–499/mo tiers)
- **Next:** receipted MRV per-deployment contracts; BYOD fleet compounds receipts
- **Later:** EcoCell hardware via crowdfunding (Crowd Supply filed), mutual-credit settlement margin

Revenue doesn't wait on hardware capital — the evidence layer is the product; devices are distribution.

Market

- **Biodiversity credits & MRV** — integrity demand rising; verifiable monitoring is the missing substrate
- **Conservation technology** — bioacoustics and remote sensing at $600+/unit vs our $35–45 BOM
- **Verification-as-a-service** — wetware compute, edge-AI efficiency claims, scientific reproducibility tooling
- **Insurance / regulatory** — audit-grade environmental evidence

Team & the honest gap

- **Thomas Price** — founder, Treetable LLC (Colorado, SAM.gov-registered)
- Built the full stack solo: Guile evidence layer, MicroPython firmware, KiCad hardware, connectome benchmark harness, live worker API
- **Gap:** solo founder, pre-field-deployment. That's part of why we're applying — program support, cofounder network, and pilot partners are what unlock the fleet stage.

Ask

- **Pre-seed / program investment** — verification services fund the first-article board run; the Crowd Supply campaign funds the fleet
- **Pilot partners** — conservation orgs, bioacoustics labs, MRV platforms that need audit-grade receipts
- **The wedge:** nobody else sells computation you can re-execute. We're the verification layer for an industry that can't prove its claims.

terexmaps@gmail.com · github: living-resources (AGPL) · compute API live today